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6.2 Special Right Triangles

Two right-triangle shapes recur so often that their side ratios are worth memorizing. A 45-45-90 **triangle** is an isosceles right triangle: if each leg is x, the Pythagorean Theorem gives hypotenuse = $x\sqrt{2}$. A 30-60-90 **triangle** comes from bisecting an equilateral triangle: if the equilateral side is $2x$, the altitude splits it into two right triangles with short leg x (opposite 30°), long leg $x\sqrt{3}$ (opposite 60°), and hypotenuse $2x$ (the original side). Once you recognize either pattern from a single given length, every other side follows immediately — no need to re-derive the Pythagorean Theorem each time. This worksheet derives both patterns and then applies them to triangles, squares, and equilateral triangles.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Legs = 9. Find the hypotenuse.

2. Hypotenuse = $6\sqrt{2}$. Find each leg.

3. Short leg = 4. Find the long leg and hypotenuse.

4. Hypotenuse = 18. Find both legs.

5. Square side 5. Find the diagonal.

6. Long leg = $5\sqrt{3}$. Find the short leg and hypotenuse.

7. Side 8. Find the altitude.

8. Hypotenuse = 14. Find each leg.

9. Short leg = $\sqrt{2}$. Find the hypotenuse.

10. Leg = $2\sqrt{5}$. Find the hypotenuse.

11. Hypotenuse = $2\sqrt{3}$. Find the short and long legs.

12. Diagonal = $10\sqrt{2}$. Find the side length.

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13. Equilateral triangle side $= 12$. Find the altitude and the area.

14. A rectangle's diagonal is 16 cm and makes a $60°$ angle with the shorter side. Find both side lengths.

15. A 30-60-90 triangle has short leg 6. Its long leg is the hypotenuse of a 45-45-90 triangle. Find the legs of that 45-45-90 triangle.

✓ Answer Key

1. $9\sqrt{2}$
2. 6
3. $4\sqrt{3}$; 8
4. 9; $9\sqrt{3}$
5. $5\sqrt{2}$
6. 5; 10
7. $4\sqrt{3}$
8. $7\sqrt{2}$
9. $2\sqrt{2}$
10. $2\sqrt{10}$
11. $\sqrt{3}$; 3
12. 10
13. Altitude $= 6\sqrt{3}$; area $= 36\sqrt{3}$
14. 8 cm and $8\sqrt{3}$ cm
15. $3\sqrt{6}$ each